System 137 (2026) 103932



Contents lists available at ScienceDirect

System

journal homepage: www.elsevier.com/locate/system





EFL learners' flow experience in story continuation writing

Yanting Wu *, Lin Jiang

Faculty of English Language and Culture, Guangdong University of Foreign Studies, Guangdong, PR China

ARTICLE INFO

Keywords:
Story continuation writing
Flow experience
EFL learners

ABSTRACT

Despite the recognized importance of flow experience in enhancing language learning outcomes, L2 writing remains an understudied domain where learners consistently report lower flow levels compared to other language tasks (Zuniga, 2023). This study addresses two gaps: (1) the over-reliance on quantitative approaches and non-linguistic theoretical frameworks in existing flow studies, and (2) the less attention given to flow in writing contexts. Focusing specifically on story continuation writing, the present study employed a qualitative analysis of 108 university students' reflective diaries to examine the phenomenological characteristics and influencing factors of flow experience. Three features of flow experience emerged from the data: transcendence of self, complex emotions, and a flood of creative flow. The study further reveals that inherent task characteristics, such as accessible linguistic models, interestingness of story, clues, and suspense setting, to name just a few, effectively stimulate learners' interest and then help learners enter the flow state. Furthermore, balancing skills and challenges through environmental control and mastery of preceding texts' languages and clues, goal-setting through outlining, and immediate feedback through creative breakthroughs also significantly affect flow experience.

1. Introduction

Writing is widely regarded as one of the most challenging skills for students to master. Compared with other tasks, learners show a relatively low level of flow experience in writing tasks (Zuniga, 2023). This phenomenon warrants attention, as flow—a psychological state characterized by complete immersion, intense focus, and autotelic experience—has been identified as a critical catalyst for optimal learning outcomes (Csikszentmihalyi, 1990). In language learning contexts, flow not only enhances proficiency by fostering engagement (Dörnyei & Ushioda, 2011) but also mitigates anxiety through positive emotional experiences (Philp & Duchesne, 2016).

Given these challenges, it is essential to explore effective methods that can enhance students' L2 writing flow experience. Among various writing tasks, story continuation writing has demonstrated unique potential. It is a reading and writing integrated task requiring learners to read a story with omitted endings and then use their imagination to complete it (Wang, 2012), The contribution of story continuation writing task to language learning has gained broad recognition (Bui & Luo, 2021; Cai & Huang, 2023; Chen & Zhang, 2022; Cui et al., 2022; Meng & Yin, 2023; Peng et al., 2020; Ren & Ma, 2024; Wang et al., 2022; Wang & Wang, 2015; Zhang, 2017, 2025; Zhang et al., 2022; Zhang & Wang, 2022; Zhou & Wang, 2024). It has been shown that the alignment effect (Wang & Wang, 2015) in the continuation writing task can significantly promote lexical learning (Chen & Zhang, 2022; Wang et al., 2022; Zhang et al., 2022; Zhang & Wang, 2022; Zhou & Wang, 2024) and enhance writing performance (Bui & Luo, 2021; Cui et al., 2022; Meng & Yin, 2023; Peng et al., 2020; Ren & Ma, 2024; Wang et al., 2022; Zhang, 2017, 2025). A meta-analysis of the alignment effect

* Corresponding author.
E-mail addresses: 513580683@qq.com, wytgdufs@outlook.com (Y. Wu), jlin@gdufs.edu.cn (L. Jiang).

https://doi.org/10.1016/j.system.2025.103932
Received 16 April 2025; Received in revised form 24 August 2025; Accepted 22 November 2025
Available online 24 November 2025
0346-251X/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

in continuation tasks (Ren &; Lu, 2021) reported a medium effect size (g = 0.581), confirming the efficacy of continuation writing in promoting language learning. Beyond boosting language learning, continuation writing also enhances learners' writing experience and cognitive development, such as stimulating learners' imagination (Wang, 2021), generating positive emotions (Jiang & Zhan, 2021; Zhang & Zhang, 2021), and reducing writing anxiety (Zhan et al., 2022). These observed effects align closely with the cognitive and affective prerequisites for flow, implying that story continuation writing may serve as an optimal context for investigating the mechanisms of flow in writing tasks.

Research on flow experience in foreign language learning has primarily focused on two strands: one compared different tasks to examine task characteristics that facilitate flow (e.g., Cho, 2018; Zuniga, 2023) or evaluates flow levels within digital foreign language learning tools (e.g., Li et al., 2021; Liu & Song, 2021), while the other investigated the relationship between flow and other psychological constructs such as self-efficacy, enjoyment, and intrinsic motivation (e.g., Chen, 2014; Dewaele & MacIntyre, 2024). These studies provide valuable insights; however, two critical gaps persist. First, the overreliance on quantitative methodologies and frameworks from non-linguistic domains risks overlooking the unique cognitive-affective features of flow in writing contexts. The flow model was developed by synthesizing the features that people exhibit when entering flow from a wide range of activities, such as sports, arts, science, and others (Csikszentmihalyi, 1990), making it highly general. Yet, within each domain, or even across different tasks, flow may manifest distinct and unique features (Almukhaild & King, 2024; Sparling & MacIntyre, 2024). Therefore, qualitative research is still needed to elicit richer and more integrated descriptions of flow experience in L2 learning tasks (Csikszentmihalyi, 2014).

Second, while oral and vocabulary tasks have received substantial attention, writing tasks—particularly creative writing like story continuation writing—remain underexplored. To address these gaps, this study aims to explore the features of flow experience and its influencing factors within the story continuation writing task to enrich domain-specific flow theories and inform pedagogical design. The present study addresses two research questions.

1. Do Chinese EFL learners experience flow in continuation writing? If so, what features of flow experience do they report?
2. In light of Csikszentmihalyi's flow theory, what specific factors affect the flow experience of Chinese EFL learners in story continuation writing?

2. Literature review

2.1. Story continuation writing

Story continuation writing is a creative writing task commonly employed in L1 or L2 contexts (Reichelt, 2009, pp. 183–206). In China, story continuation writing is an important component of the writing section in Gaokao, a highly regarded annual assessment of English proficiency for millions of Chinese university candidates. In story continuation writing, students are required to read an unfinished story and continue it by exerting their imagination (Wang, 2012). Story continuation writing holds great potential for facilitating a state of flow because it allows EFL learners to think critically and unleash their creativity (Wang, 2012). The flow experience is most likely to arise from a range of creative activities, including drawing, music composition, and creative writing (Csikszentmihalyi, 1990). Story continuation writing, which requires learners to envision the sequel to an unfinished story, relies on imagination, making it a highly creative process (Wang, 2012). Beyond the inherent creative nature of story continuation writing, previous research has also found that story continuation writing can stimulate learners' intrinsic motivation for expression (Wang, 2021), bring about positive emotions (Jiang & Zhan, 2021; Zhang & Zhang, 2021), and reduce writing anxiety (Zhan et al., 2022). These advantages of story continuation writing may make it easier for learners to enter a state of flow. Previous studies also highlighted a high level of learner engagement in this task. Zhang and Zhang (2021) conducted an investigation into a high school student's linguistic and affective development during long-term story continuation writing and observed that the participant became deeply engaged with the story, even attempting to find out the original version to discover the ending. Jiang and Zhan (2021) examined the dynamics of a non-English major's positive L2 self throughout long-term story continuation writing, and the participant in this study mentioned in his diaries that even while falling asleep, he remained preoccupied with story continuation and often got up to continue writing. Although these two studies did not directly focus on flow experiences, the reported learner engagement and deep immersion in story continuation writing suggest that participants may have achieved a state of flow during this task. However, both studies are case-based and haven't fully delved into the phenomenon of flow experience during story continuation or the factors that influence it. Therefore, further research is necessary to gain a more comprehensive understanding.

2.2. Flow theory

2.2.1. The concept of flow and its dimensions

Flow can be defined as an "optimal experience", characterized by intense concentration and immersion into a given task, which might lead to exceptional performance (Csikszentmihalyi, 1990). Flow is an elusive construct. Although a comprehensive list of defining characteristics that fully capture the flow experience has not yet been established, there are generally recognized core attributes associated with it. As summarized by Nakamura and Csikszentmihalyi (2014: 239), individuals in a state of flow typically exhibit the following characteristics: "intense and focused concentration; merging of action and awareness; loss of self-consciousness; distortion of temporal experience; a sense of control; autotelic experience".

Focused attention, merging of action and awareness, and loss of self-consciousness may be the defining characteristics of flow

(Nakamura & Csikszentmihalyi, 2014: 243). Individuals in flow are fully immersed in the activity with their attentional resources entirely preoccupied, unmindful of their surroundings and the passage of time (Christandl et al., 2018). As a result, their actions become spontaneous (merging of action and awareness), and there is no room for them to consider the past, the present, or the future (loss of self-consciousness). Csikszentmihalyi (1990) highlighted that losing self-consciousness doesn't mean losing one's sense of self or consciousness altogether. Instead, it simply means losing the awareness of one's identity. This temporary forgetting of our identity can be quite enjoyable and even allows us to expand or transcend our sense of self.

The loss of self-consciousness may also be accompanied by distortions in the perception of time during the state of flow. In this state, individuals may experience time passing either unusually quickly or slowly. However, this distortion in time perception is not as universal a feature in flow activities as other characteristics, as some activities require a sense of time (Csikszentmihalyi, 1990).

Control is another feature of flow, which refers to a profound sense of engagement and complete mastery in an activity (Csikszentmihalyi, 1990). In this state, individuals perceive themselves as capable of controlling and influencing the activity they are engaged in, with a clear understanding and predictability of possible outcomes.

Autotelic experience, sometimes termed as interest, intrinsic motivation, or enjoyment, captures learners' affect when they enter a state of flow (Egbert, 2003). Interest and the flow experience are closely intertwined. Interest induces flow experience when a person perceives an activity to be novel, important, and meaningful (Mitchell, 1988). However, interest is also regarded as a byproduct of flow experience (Aubrey, 2017). When learners experience flow in an activity, they often feel a sense of enjoyment and perceive the activity as inherently rewarding (Csikszentmihalyi, 2014). In such instances, interest arises spontaneously during the activity rather than being pre-existing. When individuals discover flow in an initially unappealing activity, they develop interest, which in turn motivates them to engage more deeply in the activity. Therefore, flow experience is "a powerful motivating force that extends an individual's goal and interest structure" and promotes the development of skills related to existing interests (Csikszentmihalyi, 2014: 233). Therefore, researchers often use the terms autotelic experience, interest, intrinsic motivation, and enjoyment interchangeably.

2.2.2. Flow antecedents: preconditions of flow

The emergence of flow experience cannot be separated from three major conditions, although these conditions cannot explain all the factors that cause flow experience (Csikszentmihalyi, 1990). The most important condition might be the balance between skill and challenge. When the task difficulty matches personal skill level, there is a high likelihood of achieving a state of flow. If the task difficulty exceeds personal skill, anxiety may ensue. Conversely, a surplus of skill in comparison to the task's difficulty can lead to boredom. The other two important conditions are clear goals and immediate positive feedback. Clear goals provide individuals with direction and purpose, enabling them to better concentrate their attention and deliberately shape their experiences in tasks. Immediate positive feedback allows individuals to understand their performance and progress, enabling them to adjust and improve their actions and decisions in time.

2.2.3. Flow experience in language learning

Although flow experience has been extensively studied within the fields of psychology and educational psychology, its application within the context of second language acquisition remains relatively underexplored. In recent decades, the emergence of positive psychology within language learning has sparked a renewed interest in examining the phenomenon of flow. Contemporary research efforts have predominantly focused on two key domains. Firstly, scholars have sought to assess the level of flow experience in digital learning environments and platforms (e.g., Li et al., 2021; Liu et al., 2023; Liu & Song, 2021), such as L2 dubbing software and virtual reality (VR)- or game-based vocabulary learning tools. Secondly, research has also delved into flow experiences in L2 tasks (e.g., Cho, 2018; Egbert, 2003; Zuniga, 2023). It has been found that flow experience is significantly related to perceived learning (e.g., Li et al., 2021; Liu & Song, 2021) and task performance, such as English reading performance (reading scores, amount of reading, and reading speed) (Gao, 2022), English vocabulary contest performance (Gao & Zhang, 2023), and degree of learning progress (Hong et al., 2017).

Furthermore, existing studies consistently suggest that tasks that evoke a higher level of flow tend to share common characteristics: they are meaningful, authentic, suitably challenging and novel, and provide clear goals, immediate feedback, and a sense of control (Csikszentmihalyi, 1990; Zuniga, 2023). Although the balance between skills and challenge is often cited as a critical factor in achieving flow states (Cho, 2018), it alone is insufficient to predict the occurrence of flow. Other factors that might influence flow experience include tasks' modality, degree of novelty, mode of collaboration, materials used, and task repetition (Payant & Zuniga, 2022; Zuniga, 2023). Besides task-related factors, learner factors, such as self-efficacy (Chen, 2014), affective states such as enjoyment and anxiety (Dewaele & MacIntyre, 2024), motivation (Hong et al., 2017), and proficiency level (Zuniga, 2023), have also been found to correlate with flow experiences.

However, current research predominantly employed questionnaire-based quantitative methods to investigate flow experience in foreign language learning, with limited use of qualitative methodologies which are more suitable for uncovering the intrinsic nature of flow experience. The manifestations of flow may vary across different domains and task types. Sparling and MacIntyre (2024) conducted a comparative analysis of learners' flow experiences in music and foreign language learning, revealing differences between "flow in music" and "flow in language". Their study indicated that while flow theory posits effortless concentration as a core characteristic, this feature was observed in musical contexts but absent in language learning. Almukhaild and King (2024) explored flow during language speaking tasks, identifying three key indicators: intense concentration, enjoyment, and a distorted sense of time. However, during the initial stages of tasks, some learners reported anxiety and heightened self-consciousness. These findings suggest that achieving flow states in language learning may present greater challenges compared to domains like music or sports. Such evidence underscores the necessity of domain-specific or task-specific flow research.

3. Methodology

3.1. Participants

This study involved 108 first-year university students majoring in English from a university in Guangdong Province in China. Due to the prevailing trend of gender imbalance in the English major in China, the sample comprised 28 male and 80 female students, aged 18 to 20. The majority of these students are from the mainland of China, with two students from the Hong Kong and Macao regions. Among the 108 students, 89 reported their scores on the National College Entrance Examination (Gaokao) in English. Their scores ranged from 105 to 143, with an average of 129 and a standard deviation of 8.46. Based on these scores, the English proficiency of these students covered both intermediate and advanced levels.

3.2. Instruments

The present study employed two instruments for data collection: the reading material and the reflection journal. The reading material, titled "*Chon*", was distributed to all participants following the class session. Participants were asked to read the provided story and compose a continuation, with a minimum word count of 500 words; no upper limit was specified. Another instrument used in the study was the reflection journal framework. We employed Csikszentmihalyi's (1990) core definition and characteristics of flow, that is, the state of being completely immersed and oblivious to the passage of time. To elicit descriptions of flow experiences from the learners and factors that affect flow, two questions were asked in the reflection journal: "During the process of story continuation writing, was there a moment when you were so immersed in writing that you forgot the passage of time? Please describe that moment in detail."; "What factors do you believe make you immersed in this writing task? Please share your observations and reflections." To facilitate accurate expression, reflective journals are written and responded to in Chinese. Data analysis was conducted entirely on the Chinese-language reflection journals. Any excerpts intended for quotation were translated into English when drafting the manuscript. The cited excerpts were first given a preliminary translation by Kimi AI which was developed by Moonshot AI. The AI tool served only as an efficiency aid. The first author then reviewed and revised the AI-translated draft carefully to ensure the original meaning was fully conveyed. After revision, the translated text was back-translated by a graduate student majoring in applied linguistics. The first author and the graduate student carefully compared the original reflective journals, the back-translated text, and the translated text, and jointly discussed and finalized the wording.

3.3. Procedures and data collection

Data collection was carried out by two instructors at a university in China. They taught "Intermediate English Writing" to first-year undergraduates, with course content centered on descriptive and narrative writing. The course included three after-class writing assignments, one of which was a story continuation writing task. Following their teaching plans, the instructors released the reflection journal Word document along with the reading materials for continuation writing. One week later, students submitted both their writing pieces and their reflection journals. During data collection, the first author of the present study served as a teaching assistant (TA) for one of the course instructors. Students were informed that the information they provided in the reflection journals would be analyzed by the TA (the first author) for research purposes and that submission of reflection journals was voluntary without affecting their grades. They were also notified that the data they provided would be anonymized upon publication. It has to be admitted that the role of TA might introduce an imbalance in power relations and thus might lead to bias in data collection. For instance, students in the class that the first author was responsible for (24 in total) may have been inclined to offer more positive or socially desirable responses in their reflection journals. However, the students well understood their contribution to the study and treated the journals as a way to interact with both the course instructors and the TA to promote their learning. They have openly discussed their writing difficulties and offered constructive and critical suggestions for improving the writing assignment, which indicates that they engage in the reflection with great candor and sincerity.

3.4. Data analysis

The responses to two questions in the reflection journal were first collated into two separate documents, which were then analyzed and coded individually. Both documents were coded using the thematic analysis (Braun & Clarke, 2006) with a combination of inductive and deductive methods. Initially, the texts were carefully and repeatedly read through to conduct open coding, with descriptions related to the feeling of entering flow and factors that influence flow being identified. To minimize the potential influence of researchers' personal biases on data analysis and make open codes as descriptive and non-judgmental as possible, open codes were assigned by adopting verbatim excerpts from the original narrative texts in the reflection journal (Hadley, 2017). For instance, the statement "when describing tension and fear, I experience those same emotions myself." gives rise to the open code "experience those same emotions myself".

Subsequently, frequency counts of similar open codes were conducted, and these codes were grouped into sub-themes. These sub-themes were compared, refined, and revised multiple times until they were mutually exclusive. For example, open codes capturing learners' momentary detachment from their physical surroundings and complete absorption into the story's scenes, such as "felt as if I could see the story's setting," "my mind was filled with images of scenes and characters," and "a scene appeared in my head", were grouped into a category labelled "cognitive transcendence". Similarly, open codes describing learners' emotional resonance with the

protagonist, such as "felt the protagonist's emotions," "experienced the same feelings myself," and "lived through her breakdown and helplessness", were grouped and labelled "affective transcendence". It is important to emphasize that although we calculated the frequency of open codes under each category, frequency counts served only as a supplementary tool to identify patterns, as high-frequency codes are not necessarily meaningful, and low-frequency ones are not necessarily irrelevant. Specifically, high-frequency codes that merely repeated question prompts of the reflection journal were excluded. Low-frequency codes underwent discussion with fellow researchers to determine their retention. Once all open codes had been grouped into sub-themes, the established sub-themes were then reclassified into larger themes using a deductive approach guided by flow theory (Csikszentmihalyi, 1990). However, the emergence of new themes not fully covered by the theory was allowed. By combining inductive and deductive methods, we were able to present the features of flow experience and the factors influencing it more comprehensively. To ensure coding credibility, the first author presented the entire coding process, including the generation of open codes, sub-themes, and themes, to an expert in applied linguistics and several fellow researchers for discussion, and then revised accordingly.

4. Results

4.1. What features of flow experience do Chinese EFL learners report in story continuation writing?

The first research question asks whether there was a moment when you were so immersed in creation that you forgot the passage of time. Of the 108 students, 99 reported experiencing such moments, 7 reported not experiencing them, and 2 provided unclear responses regarding their experience. Table 1 summarizes the phenomenological characteristics of flow experience in story continuation writing.

4.1.1. Transcendence of self

When participants entered a state of flow during story continuation writing, they often reported an inability to perceive their surroundings and a sense of losing their own identity. On one hand, they transcended themselves cognitively. Their minds were entirely occupied by the scenes of the story, and they felt like they were truly seeing the scenes in the story. They could not perceive their actual surroundings; instead, they were completely immersed in the environment of the story. On the other hand, they also transcended themselves emotionally. They deeply identified with the protagonist, and they saw what the protagonist saw and felt what the protagonist felt as if they were the protagonist themselves. Some learners even reported a strong connection with the protagonist and would like to project their personal experience and reflections onto the character due to shared life experiences. For example, in the following excerpts, the bolded texts are phrases or sentences that directly reflect that they have experienced transcendence of self.

EP1 When I feel my thoughts are clear while writing, I forget the passage of time. It feels as if everything around me ceases to exist. I become immersed in the scenes of the story, **experiencing the protagonist's joys, sorrows, anger, and happiness,** and **the scenes come to life in my mind.**

EP2 When I write about the protagonist's inner thoughts and struggles, I immerse myself in the story. For example, **when describing tension and fear, I experience those same emotions myself.**

EP3 I can't really pinpoint the exact moments when I was deeply immersed, but I was constantly trying to put myself into the story. **Especially when it came to describing the protagonist's actions, I would even act out small movements in my seat before choosing the right words.**

EP4 When writing about the protagonist's fantasies of life after getting the wealth, **I projected my own feelings into it. This is an article that reflects my profound emotions,** and it was more about drawing from my own life experiences and emotional journey to shape the story.

Additionally, while most participants mainly identified with the protagonist, they occasionally took on other identities. Some participants described feeling like the creators of the fictional world in the story, whereas others felt as if they were accomplished writers or novelists. For example.

EP5 : At that time, I was completely immersed in the story and the act of creation. It felt as if the story was real and **I was the creator of that world.**

Table 1
The phenomenological characteristics of flow in story continuation writing.

Themes	Sub-themes	Open-codes
Transcendence of self (47)	Affective transcendence (30)	e.g., experienced those same emotions myself
	Cognitive transcendence (17)	e.g., It felt like truly seeing the scenes in the story.
Complex emotion (51)	Positive high arousal emotion (34)	e.g., extremely excited; can't wait to write them down
	Positive low arousal emotion (11)	e.g., felt relaxed and satisfied
	Negative high arousal emotion (6)	e.g., struggling, painful
Creative flow (34)	Inspiration burst (24)	e.g., mind became active; with many ideas coming to me
	Smooth writing (10)	e.g., felt the words come out smoothly

Note: the number in parentheses indicates the number of references.

EP6 Despite being constantly bitten by mosquitoes, I was still able to focus on my writing. After finishing, my body was covered with itchy bites. But the more I wrote, the more energized I felt. My mind grew more active, and I even began to enjoy the process, **feeling like a great writer.**

4.1.2. Complex emotions

In the reflection journals, students experienced a range of complex emotions when they entered the flow state during the continuation writing task. Based on frequency, the following word cloud (Fig. 1) illustrates the emotions experienced by learners when entering a flow state during story continuation writing.

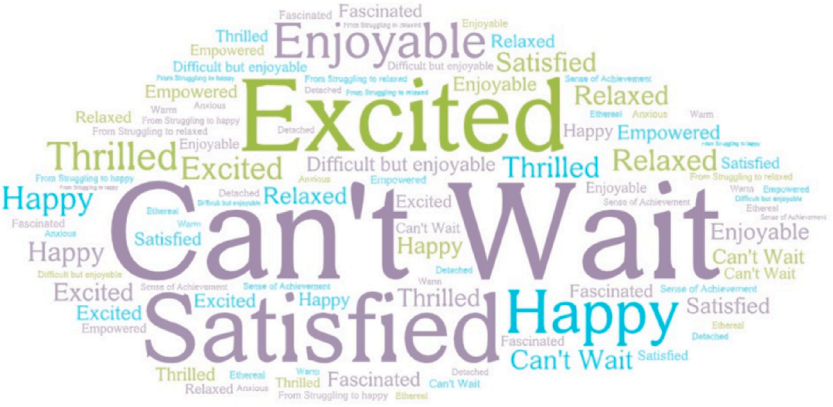
The most frequently mentioned emotions were positive high-arousal emotions, such as "can't wait (reflecting urgent expression desire)" (8), "excited" (6), "happy" (5), "enjoyable" (5), and "thrilled" (4). Interestingly, some students also mentioned a few less common and more nuanced or complex emotions, such as "detached" (1), "ethereal" (1), "fascinated" (2), and "empowered" (2). They reported that when entering the flow state, they experienced a surge of inspiration, which made them extremely excited, thrilled, and happy. At this moment, they were eager to immediately write down all the ideas in their minds. However, their typing speed or language expression ability could not keep up with the flood of inspiration. As a result, while they felt excited and happy, they also felt a bit anxious, fearing that their brilliant ideas might slip away if they didn't write them down quickly enough.

EP7 New ideas suddenly flood into my mind while I am writing. I am **extremely excited** and **can't wait to write them down.**
EP8 **My fingers are flying over the keyboard, eager to express what I am thinking, but I always feel that I am typing too slowly.**
EP9 At that moment, I am more focused on grasping the logic of the content of the article, and I am **excited and thrilled.**
EP10 As soon as I come up with the plot of the writing, I immediately pick up my pen and start writing. At this time, **I feel very excited and want to pour all my ideas into the article.**

The second most frequently mentioned emotions were positive low-arousal emotions, such as satisfied (6), relaxed (3), sense of achievement (1), and warm (1). Participants reported feeling comfortable when they had a continuous flow of inspiration. They felt satisfied and relaxed after writing down the scenes and ideas they had in mind. Some students also mentioned feeling comfortable and warm when describing warm and touching scenes.

EP11 As I was typing, a scene suddenly popped into my mind. After finishing the sentence I was working on, I inserted the scene into a suitable place in the text. During this process, my only focus was on expressing that scene. **Once I had described it, I felt a sense of satisfaction and excitement.**
EP12 It was when I was describing Mary's family going out to play. Because it corresponded with the original text and the vivid imagery had already formed in my mind, **I felt very comfortable and wrote in a relaxed and enjoyable manner.**
EP13 When I was brainstorming, I was completely immersed in the act of creation. The ideas just kept flowing endlessly, which made me **feel very comfortable and satisfied.**
EP14 When I reached the climax of the writing, my mind was teeming with ideas that I wanted to put down immediately. **This made me feel relaxed and satisfied.**

Although the majority of participants reported positive emotions, some participants also indicated that they had experienced negative emotions, such as struggling (2), painful (1), difficult (2), and anxious (1). Respondents mentioned that they would experience these negative emotions when they were out of ideas or unable to express themselves in language. Interestingly, however, after going through these negative emotions, when their thoughts became clear again, they would experience positive emotions such as happiness, enjoyment, a sense of achievement, and satisfaction. Nevertheless, some participants also noted that they felt a mixture of


Excited
Can't Wait
Satisfied
Happy
Enjoyable
Thrilled
Relaxed
Empowered
Fascinated
Difficult but enjoyable
Can't Wait
Excited
Satisfied
Happy
Relaxed
Thrilled
Enjoyable

pain and enjoyment throughout the entire process.

EP15 The emotional state at that time **transitioned from struggle to clarity and relaxation.**
EP16 I felt I had been immersed in the characters and the scenes of the story, with a continuous flow of inspiration. There was **a mix of excitement** (as the ideas started coming), **urgency** (wanting to finish the assignment quickly), **and a bit of pain** (fearing that the inspiration would disappear and leave me stuck).
EP17 At first, right after reading, I was really confused and had no idea what to write. I was **stuck there for quite a while**. But **when I finally came up with the content for the continuation, I felt extremely happy.**

4.1.3. Creative flow

The last typical phenomenological characteristic of entering a state of flow is that new ideas keep emerging, and the writing direction is extremely clear and smooth. In this state, they felt that their minds were suddenly enlightened and highly active, and they were also very clear about what the next plot would be.

EP18 The emotional state at that time transitioned from struggle to clarity and relaxation. **My mind was quite active, with ideas emerging one after another,** and my brain was filled with images of the imagined characters.
EP19 I felt as if I had been immersed in the characters and the scenes of the story, **with a continuous flow of inspiration.** At that moment, **my mind was very active** and I was quite excited.
EP20 When I reached the climax of the writing, **my mind suddenly became particularly active**, with **many ideas coming to me**. The writing speed increased, and my mood was quite good.
EP21 I felt like I had **truly experienced what inspiration is all about.** No wonder writers are always in search of it. Moreover, **my mind was quite active. While writing halfway through, I would already have the ending description in mind.**

4.2. What factors do Chinese EFL learners believe affect their flow experience?

4.2.1. Interest

When asked about the factors that led them into a state of flow, the frequently mentioned factor was their interest and enthusiasm for this writing task (34 references). On the one hand, the story itself was highly engaging (13 references), which allows them to immerse in the story and have a strong sense of involvement (10 references). On the other hand, the clues and suspense (8 references) in the preceding text sparked their imagination and desire to write. A small number of participants also mentioned that their inherent interest in story creation (3 references) contributed to their entering a state of flow.

EP22 **The free and unrestricted imaginational space** leading me to the state of the state. It's **the clue** in the original story that drives me into such an absorbed condition.
EP23 **Imagination**, imagining oneself being at the scene or even **stepping into the shoes of the characters**, can better complete the writing.
EP24 **The original text's details and environmental descriptions** make me feel as if they are right there.
EP25 **I think it is the imagination and interest in the story.** While writing, the story occupies my mind completely. What I think about is how to express the scenes that emerge in my mind.
EP26 **The questions left by the story itself** and the burst of creative inspiration.
EP27 **I think it should be the exploration of the story's details.** Because I need to base on the previous text and the clues I have discovered. Combined with my own imagination, I construct the overall direction of the story and the specific plot. This puts me into a state of flow.

4.2.2. The balance between skill and challenge

When the balance between skill and challenge (35 references) was achieved, learners were more likely to enter a state of flow. Environmental control was frequently mentioned as an important element. A quiet environment (21 references) can make it easier for them to immerse themselves in the story.

EP28 **The quiet atmosphere of the library** provides me with the space for deep thinking and an immersive environment for writing.
EP29 **A quiet, comfortable, and interference-free environment.** Such an environment is favorable for focusing one's attention.

Some participants also said that listening to music (4 references) while doing the continuation writing helped them immerse themselves in the story and stimulated their imagination.

EP30 Moreover, **I played music** during the creative process. **I felt very inspired** while writing, and **there was a kind of fateful overlap between the music and my content.**

Furthermore, the language of the preceding text was available for imitation (6 references), which reduced their output pressure. Therefore, they mentioned that becoming familiar with the plot and linguistic features of the preceding text before writing can make their writing more fluent. Additionally, some participants indicated that familiar topics and themes (4 references) also facilitate entering a state of flow.

EP31 In the early stages of writing, I always went back to the original text to confirm the author's expressions. **My familiarity with the language features of the original text in the early stage reduced my hesitation in expression later on.**

EP32 **Familiarity with the topic and theme** helped me enter this state of full concentration. Familiarity makes it easier to get into the zone, think more quickly, and start creating.

4.2.3. Clear goals and positive feedback

Having a clear goal (11 references) before writing, and receiving positive feedback (7 references) throughout the writing process, helped learners enter a state of flow and prolong the duration of this state. Creating an outline or a mind map before writing can make the writing process smoother. For example, some participants mentioned.

EP33 **Having a clear outline** makes me more aware of what I should do at each stage.

EP34 **The outline** prepared in advance: After reading the text, I initially determine a plot direction. When writing, I gradually develop the listed plot points, following my outline step by step. This way, I don't have to write a scene and then stop to think about what comes next, so I can stay more focused.

Moreover, during the writing process, when they were highly satisfied with their own plot design, it tended to make them more immersed. This usually happened when inspiration struck, and this positive feedback led them to experience positive, high-arousal emotions.

EP35 **I feel that the most important factor is that the idea I came up with is quite novel and interesting.** I wanted to write the essay as if I were creating a piece of art, and that's why I easily got immersed and enjoyed the process.

EP36 **I think it's because I have confidence in the story I set up**, and the motivation to see it completed drives me to keep writing.

EP37 **The main reason is the desire to write which is sparked by the interesting plot I created.** When I was writing, I thought **the setting I came up with was quite interesting**. I am **satisfied with it,** although it's not a particularly innovative type. If I can continue the story with appropriate language and plot, the story would still be smooth and reasonable.

5. Discussion

The first research question reveals the phenomenological characteristics of flow experience in story continuation writing. From the data, it can be concluded that when entering a state of flow during story continuation writing, individuals experienced transcendence of self, complex emotions, and a flood of creative flow. Firstly, after entering flow during the continuation writing, people no longer had a sense of their own identity. At this time, they immersed themselves in the world of the story, where they might be the protagonist, the creator of that world, or even a great writer working on a masterpiece. This finding is consistent with previous research (Csikszentmihalyi, 1990; Sparling & MacIntyre, 2024), which suggests that when individuals reach an optimal flow experience, they sometimes transcend themselves. That means all their attentional resources are focused on how to depict the story well, with no room to worry about their past, present, or future. This temporary forgetting of one's identity allows them to escape from worldly troubles, which is one of the sources of the intense happiness that people derive from entering flow (Csikszentmihalyi, 1990). Secondly, after entering flow, learners experienced a range of positive emotions. This finding validates Csikszentmihalyi's (1990) theory that flow experiences can bring about positive emotions and arouse intrinsic motivation. This is why flow experiences are often closely linked with positive emotions (Dewaele et al., 2025; Dewaele & MacIntyre, 2024) and intrinsic motivation (Hong et al., 2017). However, not every moment in the continuation writing process is a flow state. Sometimes they experienced excitement and anxiety at the same time in the flow state. Some learners who have many ideas after reading the preceding text may directly experience positive high-arousal emotions such as excitement and enjoyment. Other learners may initially be in a state of struggling and anxiety while also enjoying the process, and they would feel a sense of satisfaction and accomplishment after completing the task. Research by Almukhaild and King (2024) also indicated that the emotions experienced during language learning tasks are highly complex. Lastly, learners reported that entering flow is often accompanied by clear thinking and a surge of inspiration. This finding suggests that when learners have a clear goal in their writing (clear thinking) and receive positive feedback (surge of inspiration), they are likely to enter a state of flow. The results are consistent with previous studies (e.g., Li et al., 2021; Liu et al., 2023; Liu & Song, 2021) which have shown that when entering a state of flow, learners always know what they need to do next and how well they are performing at each step. Therefore, clear goals and immediate positive feedback are not only phenomenological characteristics of flow but also considered the prerequisites for entering a state of flow (Csikszentmihalyi, 1990).

The second research question reveals factors that affect the flow experience. This study found that stimulating interest, balancing skill and challenge, having clear goals and positive feedback are essential in entering flow experience. Firstly, the inherent advantages of continuation writing tasks facilitated flow through stimulating learners' interest. The clues and suspense in the preceding text stimulated the learners' imagination, curiosity, and their intrinsic motivation before they began writing, and the interesting story allowed learners to temporarily forget their own identities and step into the roles within the text. The findings corroborate previous research, which indicates that continuation writing can elicit intrinsic motivation for expression (Wang, 2021), generate positive emotions (Jiang & Zhan, 2021; Zhang & Zhang, 2021). The findings also demonstrate that interest and flow experience are interrelated (Csikszentmihalyi, 2014). On the one hand, pre-existing interest in a certain task makes it easier for individuals to enter a state of flow (Mitchell, 1988). On the other hand, interest can also arise spontaneously during the task rather than being present from the beginning (Aubrey, 2017). Additionally, the preceding text in the continuation writing task plays a role in balancing skill and challenge by

reducing learners' writing anxiety (Zhan et al., 2022), making the writing process smoother and reducing the time spent agonizing over language expression. In addition to the influence of the preceding text, having a clear writing outline before writing also helped learners enter a state of flow. Once in flow, the emergence of inspiration, as a form of positive feedback, continuously drove learners toward the peak of flow experience. These findings once again confirm that the balance between skill and challenge, clear goals, and positive immediate feedback are prerequisites for entering a flow experience (Csikszentmihalyi, 1990).

6. Conclusion, implications, and limitations

This study investigated the phenomenological characteristics of flow in story continuation writing and the factors that influence the flow experience. It was found that when learners entered a state of flow during story continuation writing, they experienced transcendence of self, complex emotions, and a flood of creative flow. In story continuation writing, stimulating learners' interest is crucial. For instance, increasing the interestingness of the story, setting up clues and suspense can stimulate their imagination and desire to express themselves and help them enter the flow state. Furthermore, balancing learners' skills and the challenges posed by the task, such as controlling the familiarity of the story topic and creating a quiet writing environment. Finally, clear goals and positive feedback helped maintain flow. For example, having an outline before writing helped learners enter flow. This study holds significant theoretical and practical implications. On the one hand, it provides valuable additions and insights for research on flow experiences in foreign language learning. On the other hand, it helps teachers assist learners in finding flow in writing and overcoming writing anxiety. For example, teachers can provide targeted help based on the factors that influence writing flow. They can help learners interpret clues and suspense to stimulate their imagination, or explain the background of the story and imitable language points to increase the interestingness and familiarity of the story. While the study has many important findings, it also has some limitations. First, the study relies on reflection journals, which cannot record the entire writing process in real time and fully capture the learners' instantaneous psychological changes and the specific details of their flow experiences during writing. The second limitation concerns the potential for social desirability bias in reflection journals. Students, particularly within the Chinese university context, might tend to report positive experiences to meet teachers' expectations even though they were verbally and explicitly assured that their journal entries would not influence their course performance. Third, the sample is not diverse enough, involving only English major students, and thus cannot fully reflect the differences in flow experiences among learners from different backgrounds in story continuation writing. In addition, the study did not look at the duration and frequency of flow states. Future research could employ a variety of research methods, such as eye-tracking technology, to more precisely record and analyze learners' flow experiences to address these shortcomings.

CRediT authorship contribution statement

Yanting Wu: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Lin Jiang:** Writing – review & editing, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the authors utilized ChatGPT and KIMI for proofreading and polishing. All suggested revisions were carefully reviewed, and selected changes were incorporated as deemed appropriate. The authors assume full responsibility for the content of the publication.

Funding

This work was supported by the Guangdong Provincial Department of Education's International Training Program for Outstanding Young Researchers and Guangdong University of Foreign Studies.

Declaration of competing interest

None.

Acknowledgments

The authors wish to express their profound gratitude to Professor Yinyin Du, Professor Hongying Peng, and all the participants for their invaluable support during data collection. We are also grateful to Professor Feng Ding and our colleagues for their insightful suggestions during data analysis.

References

Almukhaild, H., & King, J. (2024). Scaling the apex of engagement: Anxious language learners' positive flow experiences during speaking tasks. *System, 124*, Article 103373.

Aubrey, S. (2017). Inter-cultural contact and flow in a task-based Japanese EFL classroom. *Language Teaching Research, 21*(6), 717–734.

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology, 3*(2), 77–101.
Bui, G., & Luo, X. Y. (2021). Topic familiarity and story continuation in young English as a foreign language learners' writing tasks. *Studies in Second Language Learning and Teaching, 11*, 377–400.
Cai, D. H., & Huang, L. Y. (2023). Effects of input modality on alignment in continuation writing. *Journal of Second Language Writing, 62*, Article 101060.
Chen, P. H. (2014). *The relationship of flow experience, need satisfaction, perceptual learning style preferences, and EFL self-efficacy to EFL online learner satisfaction (doctoral dissertation*. Washington State University.
Chen, J. H., & Zhang, X. P. (2022). L2 development of phraseological knowledge via a xu-argument based continuation task: A latent curve modeling approach. *System, 106*, Article 102767.
Cho, M. (2018). Task complexity and modality: Exploring learners' experience from the perspective of flow. *The Modern Language Journal, 102*(1), 162–180.
Christandl, F., Mierke, K., & Peifer, C. (2018). Time flows: Manipulations of subjective time progression affect recalled flow and performance in a subsequent task. *Journal of Experimental Social Psychology, 74*, 246–256.
Csikszentmihalyi, M. (1990). *Flow: The psychology of optimal experience*. New York: HarperPerennial.
Csikszentmihalyi, M. (2014). *Flow and the foundations of positive psychology*. Dordrecht: Springer.
Cui, Y. Y., Yang, L. R., & Wolter, B. (2022). Alignment effect in the continuation task of Chinese low-intermediate English learners. *Applied Linguistics Review, 13*, 501–526.
Dewaele, J. M., & MacIntyre, P. (2024). "You can't start a fire without a spark". Enjoyment, anxiety, and the emergence of flow in foreign language classrooms. *Applied Linguistics Review, 15*(2), 403–426.
Dewaele, J. M., Macintyre, P., Kamal Ahmed, I., & Albakistani, A. (2025). Emotional, attitudinal, and sociobiographical sources of flow in online and in-person EFL classrooms. *Applied Linguistics, 46*(1), 91–109.
Dörnyei, Z., & Ushioda, E. (2011). *Teaching and researching: Motivation* (2nd ed.). Harlow: Longman.
Egbert, J. (2003). A study of flow theory in the foreign language classroom. *The Modern Language Journal, 87*(4), 499–518.
Gao, Y. L., & Zhang, Z. Y. (2023). Flow and anti-flow in college English contests: Antecedents and predictive effects. *Foreign language world, 4*, 64–72.
Gao, Y. L., Zhang, Z. Y., & Ni, C. B. (2022). Flow experience in foreign language reading and its relationship with reading performance. *Foreign language world, 3*, 70–78.
Hadley, G. (2017). *Grounded theory in applied linguistics research: A practical guide*. Abingdon: Routledge.
Hong, J. C., Hwang, M.-Y., Tai, K. H., & Lin, P. H. (2017). Intrinsic motivation of Chinese learning in predicting online learning self-efficacy and flow experience relevant to students' learning progress. *Computer Assisted Language Learning, 30*(6), 552–574.
Jiang, L., & Zhan, J. (2021). A case study of positive L2 self-development in the iterative continuation task of writing. *Foreign Language World*, (6), 23–30.
Li, R., Meng, Z., Tian, M., Zhang, Z., & Xiao, W. (2021). Modelling Chinese EFL learners' flow experiences in digital game-based vocabulary learning: The roles of learner and contextual factors. *Computer Assisted Language Learning, 34*(4), 483–505.
Liu, Y. C., Huang, T. H., & Lin, I. H. (2023). Hands-on operation with a rolling Alphabet-AR system improves English learning achievement. *Innovation in Language Learning and Teaching, 17*(4), 812–826.
Liu, H., & Song, X. (2021). Exploring "Flow" in young Chinese EFL learners' online English learning activities. *System, 96*, Article 102425.
Meng, L., & Yin, H. (2023). Effects of input enhancement and genre on L2 learners' performance in the continuation writing task. *Humanities and Social Sciences Communications, 10*, 959.
Mitchell, R. (1988). Sociological implications of the flow experience. In M. Mitchell, & I. Csikszentmihalyi (Eds.), *Optimal experience: Psychological studies of flow in consciousness* (pp. 36–59). Cambridge: Cambridge University Press.
Nakamura, J., & Csikszentmihalyi, M. (2014). The concept of flow. In *Flow and the foundations of positive psychology: The collected works of Mihaly Csikszentmihalyi* (pp. 239–263). Netherlands: Springer. Dordrecht.
Payant, C., & Zuniga, M. (2022). Learners' flow experience during peer revision in a virtual writing course during the global pandemic. *System, 105*, Article 102715.
Peng, J., Wang, C., & Lu, X. (2020). Effect of the linguistic complexity of the input text on alignment, writing fluency, and writing accuracy in the continuation task. *Language Teaching Research, 24*, 364–381.
Philp, J., & Duchesne, S. (2016). Exploring engagement in tasks in the language classroom. *Annual Review of Applied Linguistics, 36*, 50–72.
Reichelt, M. (2009). A critical evaluation of writing teaching programmes in different foreign language settings. *Writing in foreign language contexts: Learning, teaching, and research*.
Ren, W., & Lǔ, X. X. (2021). Meta-analysis of the alignment effects in the continuation writing task. *Foreign Language World, 4*, 44–52.
Ren, W., & Ma, D. (2024). Incorporating iterative continuation tasks into EFL writing courses. *Language Teaching Research*, Article 13621688241286734.
Sparling, H., & MacIntyre, P. (2024). A tartan weave: Connecting the experience of flow in traditional music and gaelic language in pursuit of heritage language survival. *Journal of Multilingual and Multicultural Development, 45*(1), 47–58.
Wang, C. (2012). Use of the continuation task to improve L2 learning efficiency. *Foreign Language World, 5*, 2–7.
Wang, C. (2021). The rationale for the xu-argument's high efficiency of L2 learning. *Foreign Language World*, (6), 2–7.
Wang, M., Gan, Q., & Boland, J. E. (2022). Effect of interactive intensity on lexical alignment and L2 writing quality. *System, 108*, Article 102847.
Wang, C. M., & Wang, M. (2015). Alignment effect on L2 written production. *Applied Linguistics, 36*, 503–526.
Zhan, J., Jiang, L., & Huang, L. (2022). An investigation of L2 writing anxiety in the comparative continuation task. *Journal of PLA University of Foreign Languages, 45*(2), 61–69.
Zhang, X. Y. (2017). Reading–writing integrated tasks, comprehensive corrective feedback, and EFL writing development. *Language Teaching Research, 21*, 217–240.
Zhang, H. (2025). Effect of interaction and alignment of continuation task in Chinese as a second language writing. *Journal of Language, Identity and Education, 24*(1), 1–13.
Zhang, X. Y., Du, L., & Zhu, X. Y. (2022). Effects of the continuation task and the RCC task on Japanese as a foreign language vocabulary learning. *Applied Linguistics, 43*, 725–745.
Zhang, X., & Wang, M. (2022). The continuation task and the model-as-feedback writing task in L2 vocabulary learning: Timing of model texts. *Language Teaching Research*, Article 13621688221119077.
Zhang, S., & Zhang, L. J. (2021). Effects of a xu-argument-based iterative continuation task on an EFL learner's linguistic and affective development: Evidence from errors, self-initiated error corrections, and foreign language learning attitude. *System, 98*, Article 102481.
Zhou, X., & Wang, C. (2024). Effects of interactive alignment on L2 vocabulary learning by Chinese EFL learners. *Language Teaching Research, 28*, 466–496.
Zuniga, M. (2023). The correlates of flow in the L2 classroom: Linking basic L2 task features to learner flow experiences. *The Modern Language Journal, 107*(3), 650–668.